Analyzing fish spatial and temporal distribution patterns using un-calibrated acoustic data from multiple vessels

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Abstract

Acoustic density data were collected from three commercial fishing vessels participating in the 2003 eastern Bering Sea walleye pollock (*Theragra chalcogramma*) fishery. Unique challenges encountered with these data included: 1) uncalibrated data from each vessel, 2) no formal sampling design, 3) data volumes that exceeded computing capacity, 4) unbalanced temporal and special coverage, 5) a high degree of spatial autocorrelation, and 6) heteroskedastic density distribution between light and dark hours. We assess spatial and temporal structure of pollock and present a stratified random sampling technique for subsampling the data and addressing unbalanced data collection, while retaining the spatial and temporal attributes of the pollock distribution within the data. A general additive mixed model (GAMM) with an internal correlation structure and heterogeneous variance was used to inter-calibrate the data and to evaluate spatial and temporal distribution of pollock density on the eastern Bering Sea shelf. Covariates investigated included vessel, daylight phase, week of the year, bottom depth, and sea surface temperature. Initial optimum GAMM configuration was evaluated using small sample Akaike information criterion (AIC_c). Resampling cross-validation is presented as a means to counter possible over-parameterization of smoothing splines where a GAMM was not initially optimized over the full range of data. This study introduces a new data source and presents analytic methods to quantify factors influencing fish distributions and dynamics at spatial and temporal scales that were not previously possible.

Keywords: Opportunistic acoustic data, Alaska pollock, unbalanced subsample, inter-calibration, generalized additive mixed model, temporal-spatial analysis

Introduction

In 2002 scientists at the Alaska Fisheries Science Center in collaboration with the University of Alaska, Fairbanks, and members of the commercial fishing industry with funding through the Pollock Conservation Cooperative Research Center began collecting acoustic data from multiple commercial fishing vessels during normal fishing operations in the eastern Bering Sea walleye pollock (*Theragra chalcogramma*) fishery. This collection has continued through the present day. This opportunistic data collection provides an inexpensive means to acquire data from multiple vessels over a broader range of time scales than is usually collected during dedicated systematic surveys. A multiple platform approach potentially offers a more synoptic view of a population than a survey conducted by a single vessel and provides data at a wide range of temporal and spatial scales from meters to hundreds of kilometers and minutes to months. The intended use of these data is to investigate possible environmental and fishing effects on pollock density and distribution, but due to the unorthodox manner in which these data are collected a number of issues must be addressed before they can be used quantitatively for this purpose.

The application of opportunistic acoustic data to scientific inquires is highly problematic because of the large volume of data collected, the lack of system and inter-vessel calibrations, and the absence of a predetermined sampling design. This study proposes methods for using these data to investigate the distribution and dynamics of walleye pollock on the eastern Bering Sea shelf during the winter fishery. Although these data do not meet the standards suggested by ICES (2007), we propose that opportunistically collected, uncalibrated acoustic data can be used quantitatively where absolute estimates of abundance are not necessary. We apply methods developed in the knowledge discovery and data mining field to intelligently reduce the large unbalanced dataset to a size manageable in standard statistical software while retaining spatial and temporal trends of interest. Further, we demonstrate the use of generalized additive mixed models (GAMM) to standardize the data among vessels and assess the contribution of environmental covariates to pollock density and distribution. Finally, due to the size of the dataset and inability to fit all the data in a single model run, we propose the use of small sample Akaike information criterion (AIC_c) to choose among competing models configurations, then apply resampling cross-validation to determine the optimum malleability of the GAMM smoothing functions.

Methods

Data collection and processing

From 18 January through 26 March 2003 we collected acoustic data along approximately 32,000 km of trackline from hull mounted 38 kHz SIMRAD ES-60 echosounders on three factory trawlers participating in the eastern Bering Sea walleye pollock fishery (Fig.1). The data were processed using Echoview (version 3.1 SonarData Inc.). Areas of incomplete transmission (missing pings) and data judged to be of poor quality were discarded. The filtered backscatter data were integrated with a threshold value of -70db to nautical area scattering strength (S_A db re 1 (m^2nmi^{-2}); MacLennan et al. 2002) at 100 m horizontal bins extending from 15 m below the surface to 0.5 m from the bottom. All backscatter was attributed to pollock. Although this assumption is an obvious simplification, data from formal acoustic surveys conducted in the study area in February 2001 and 2002 show that ~98% of all backscatter could be attributed to pollock (Honkelehto et al, 2002).

Each sampling unit was classified by the decimal time of day, date in Alaska Standard Time (AKST), daylight/night, fishing status, sea surface temperature (SST), and location in UTC kilometers. SST data were obtained from the National Climate Data Center (NCDC, 2008). SST data have a spatial grid of 0.25° and a temporal resolution of one day and were available for the entire Bering Sea for 2003. For January through March in the Bering Sea the water column is not vertically stratified and SST provides a good representation of water column temperatures (Stabeno et al 2007). For this reason we limited our analysis to the shelf area with bottom depths less than 150 meters.

Lack of sphere calibrations

ICES Cooperative Research Report No. 287 (ICES, 2007) provides guidance for the collection and use of acoustic data from commercial fishing vessels. The report states that the use of uncalibrated data is inappropriate for quantitative purposes. Sphere calibrations are necessary to assess how measurements differ from a known reference target, to evaluate the equivalent two-way beam angle, and to assess the stability of the acoustic system over time (Foote et al, 1993). Data collected in this study do not conform to these guidelines as the acoustic systems have not been calibrated using a standard target. Without calibration, acoustic-based abundance estimates may not be accurate or precise. If measurements are consistent (i.e. changes in gain values are low) data from uncalibrated echosounders can be used to derive density or abundance indices. The nautical area scattering coefficient derived from an uncalibrated acoustic system (s'_A ; $m^2 nmi^{-2}$) is assumed proportional to the "true" nautical area scattering coefficient (s_A ; $m^2 nmi^{-2}$) from a calibrated system. The calibration offsets for the two values evaluated in a sphere calibration (two-way beam angle (Ψ ; dBre 1 Steradian) and difference from theoretical target strength (ΔTS; dB re 1 m^2)) can be expressed as the two-way beam angle correction (Ψ_{cor}) or in linear terms the equivalent two-way beam angle correction (ψ_{cor}), the correction to the area backscattering strength (S_{acor}), and the correction to the system gain (G_{cor}). These two terms were converted from logarithmic to linear terms and combined in a single correction term (*cor*). A conversion factor from scattering coefficient to mass (b_{con}) enables biomass estimates using s_A values.

$$\textit{Where} \quad \Psi = \Psi' + \Psi_{cor} = 10Log_{10}(\psi) = 10Log_{10}(\psi' \cdot \psi_{cor})$$

$$V = (r_2^3 - r_1^3)(\psi' \cdot \psi_{cor}), \quad cor = 10^{\left(\frac{(S_{acor} + G_{cor})}{10}\right)}, \quad b_{con} = \frac{a\overline{w}}{4\pi\langle\sigma_{bs}\rangle}, \quad \textit{and}$$

$$s'_A = (4\pi(1852)^2)\int_{z1}^{z2}\left(\frac{1}{\psi'(r_2^3 - r_1^3)}\right)\sum \sigma'_{bs} dz = \frac{(4\pi(1852)^2)}{\psi'(r_2^3 - r_1^3)}\int_{z1}^{z2}\sum \sigma'_{bs} dz, \quad \textit{then}$$

$$s_A = \frac{\left(4\pi(1852)^2\right)}{(\psi' \cdot \psi_{cor})(r_2^3 - r_1^3)}\left(\int_{z1}^{z2} \sum cor\,\sigma'_{bs} dz\right) = \left(\frac{cor}{\psi_{cor}}\right)\left(\frac{4\pi(1852)^2}{\psi'(r_2^3 - r_1^3)}\right)\left(\int_{z1}^{z2} \sum \sigma'_{bs} dz\right),$$

$$\therefore s_A = s'_A\left(\frac{cor}{\psi_{cor}}\right), \quad and$$

$$where \quad b = s_A b_{con}, \quad then \qquad s'_A\left(\frac{b_{con} cor}{\psi_{cor}}\right) = b,$$

$$\therefore s'_A \propto b$$

Where Ψ' (dB re 1 Steradian) and ψ' (Steradian) are the factory calibrated two-way and equivalent two-way beam angles respectively, V (m^3) is the sample volume, a (nmi^2) is the survey area, σ'_{bs} (m^2) is the backscatter cross-section from the uncorrected system, $\langle \sigma_{bs} \rangle$ (m^2) is the expected backscattering cross-section for the species of interest , $\overline{w}$ (t) is the mean weight of the species of interest, r (m) is the range from the transducer face, and z are the depths over which the analysis is conducted (Medwin and Clay, 1998; MacLennan et al, 2002; Simmons and MacLennan, 2005).

Spatial and temporal analysis

We investigated both spatial autocorrelation with Moran's I cross-correlograms (Bjornstad and Falck, 2001) as implemented in the *correlog* function provided in the **ncf** R library (ncf version 1.1-2, 2009) and temporal autocorrelation with the auto-correlation function (ACF; Venables and Ripley, 2002) as implemented in the *acf* function provided in the **stats** R library (stats version 2.8.1, 2009). Moran's I cross-correlograms were calculated for nautical area scattering strength (S_A; dB re $1m^2nm^{-2}$) at spatial scales of 0.5, 1, and 10 km for both daytime and nighttime data for each vessel. The 95% confidence intervals of the Moran's I statistic were calculated through bootstrapping the data at the particular lag for 1,000 iterations (Bjornstad and Falck, 2001). The ACF were produced for all three vessels for one hour lags out to 200 hours and 24-hour lags out to 50 days. Significant levels were set at 5% (α=0.05).

Handling large unbalanced datasets

A particularly difficult problem encountered in using these data was that the quantity of data exceeded the processing capabilities of the computer resources available to us. Due to the spatially and temporally unbalanced data collection simple random sampling under sampled areas that were not fished, but were transited through, and days in which there were few samples. We used an inverse density biased sampling (Palmer and Faloutsos, 2000; Nanopolous et al 2003) approach with the data stratified over space, time, and vessel to thin the data in strata where the effective sample numbers were much smaller than the actual number of sample points. Inverse density biased sampling (IDBS) probabilistically under-samples dense strata and oversamples lower density strata. Density biased sampling uniformly samples data points within strata while preserving sample density; uniform sampling is a special case where all strata are sampled at the same proportion, as is IDBS where proportionally more samples are selected from less dense strata. We used the formula

$$P(\textit{including point } x \mid x \in c) = f(n_c) = \frac{L}{n_c^f \sum_{i=1}^{G} n_i^{1-f}}, \quad 0 \le f \le 1,$$

where G is all strata, L is the over all sample size wanted, n_c is the number of sample points in strata c, n_i is the number of sample points in strata i, and f is a constant which sets the degree of density bias. Where $f = 0$ the resulting sample would be uniform or simple random sampling and where $f = 1$ the expected number of samples from each strata would attempt to achieve equivalence and therefore, inversely biased to the density of points in the strata. In this study we used $f = 1$ due to the highly spatially and temporally unbalanced sample. In reference to the geostatistical analyses conducted earlier, we created a spatial sampling grid at 10 km × 10 km, the approximate initial zero intercept of the spatial cross-correlograms and range of semi-variograms for all vessels. Julian day was chosen as the temporal scale over which to stratify the samples because at smaller scales the diel trend in backscatter was consistent over the season and over the entire study area. To determine an appropriate sample size and estimate the uncertainty around the subsamples we calculated the mean squared error (MSE) of the mean backscattering strength values for sample sizes between 5,000 and 100,000 over 1,000 iterations. The sample size in which the coefficient of variation of the square root of the MSE (CV_{RMSE}) equaled 10% was chosen for this study, calculated as:

$$MSE_L = \frac{1}{M_0} \sum \left(\bar{x}_{ijkL} - \bar{X}_{ijk} \right)^2 ,$$

$$CV_{RMSE} = \frac{\sqrt{MSE_L}}{\frac{1}{M_0} \sum \bar{X}_{ijk}},$$

where M_0 is the number of strata, $\bar{x}_{ijkM}$ is the mean backscatter strength of the subsampled data for the 100 km^2 spatial grid i, Julian day j, vessel k, sample size L, and $\bar{X}_{ijk}$ is the mean backscatter strength for the full dataset for each strata (Palmer and Faloutsos, 2000).

Standardization and environmental covariates

Standard inter-vessel calibrations (Monstad et al., 1992; Wyeth et al., 2000; Mason and Schaner, 2001; Simmonds and MacLennan, 2005) could not be conducted in this study. The need to standardize fisheries data among vessels is not unique to acoustic data collection. Regression models, including simple linear, mixed effect, and generalized linear and additive models (Venables and Dichmont, 2004), have been widely used to standardize catch per unit effort (CPUE) data among commercial fishing vessels (Maunder and Punt, 2004; Su et al 2008). Maunder and Punt (2004) provides an excellent overview of the use of statistical models to standardize commercial catch data for use as indices of abundance in stock assessments. With the understanding that both CPUE and acoustic backscatter are measures of population density, we employ a generalized additive mixed model (GAMM) to standardize data among vessels similar to procedure commonly used for standardizing CPUE data. In addition, we included two environmental covariates, sea surface temperature (SST) and bottom depth, in the GAMM analysis to illustrate how this methodology could be used to predict the effects of environmental covariates on relative pollock density.

For the GAMM analyses our dependent variable is uncalibrated backscattering strength (S'_A; dB re 1(m^2nmi^{-2})) and we assumed a Gaussian error distribution. We investigated vessel, daylight, week of the year, bottom depth (m), and SST (°C) as fixed independent variables. Smoothed terms were fit as isotropic thin plate regression splines (Wood, 2006a; Wood, 2006b). All analyses were conducted in R (64 bit V. 2.5.0; R Development Core Team, 2008) using the **mgcv** package (V. 1.3-31; Wood, 2007) on an HP Proliant DL 385 with dual 1800MHZ AMD Opteron Processors and 16 GB of memory operating on a Red Hat Linux Enterprise Server 4. The inverse density biased strata described above was treated as an independent random variable. Heteroscedasticity of the variance structure between the within group errors for day and night was explored within the model using the *varIdent* variance function described by Penheiro and Bates (2000). Two-dimensional spatial autocorrelation at small scales (< 10 km) was investigated in the model using an exponential correlation structure fit on UTC location to the nearest 0.01 km initialized with a range of 1.0 and nugget of 0.5 (*corExp*; Penheiro and Bates, 2000; Wood, 2006).

Six initial model configurations (Table 2) were fit with restricted maximum likelihood (REML; Wood, 2007) all smoothes in these initial runs were limited to a basis dimension (k) of four to reduce processing times. The small sample Akaike information criterion (AIC_c; Burnham and Anderson, 2002) was calculated for each model. The model configuration with the smallest AIC_c was chosen for further analysis. Because only a subsample of the data was used in the model development, cross-validation was used to select the basis dimensions over which the smoothes were fit. This in practice set the maximum degrees of freedom associated with a smooth and therefore its stiffness. For the cross-validation the best candidate model was fit using REML over a range of k values for each smooth term (Table 3). The data not used in the original model fit were re-sampled 1,000 times with replacement using IDBS selecting 1,000 data points for each iteration. The S'_A was then predicted for all of the data subsets for each model smooth configuration and the residual sum of squares (RSS_p), prediction standard error ($\hat{\sigma}$) and log-likelihood (LL_p) calculated using:

$$LL_p = -\frac{1000}{2}\log\left(\hat{\sigma}^2\right) - \frac{1000}{2}\log(2\pi) - \frac{1000}{2},$$

where $RSS_p = \sum\left(\hat{S}'_{Ai} - S'_{Ai}\right)^2$ and $\hat{\sigma} = \sqrt{\frac{RSS_p}{1000}}$,

and $\hat{S}'_{Ai}$ is the predicted uncalibrated backscattering strength (Burnham and Anderson, 2002). The model configuration with the highest proportion of highest LL_p was chosen as the preferred model.

A projection dataset covering the entire eastern Bering Sea at a 100 km^2 resolution was constructed for 20 January through 1 April 2003. Each point in the dataset included daily average sea surface temperature, bottom depth, UTC location, IDBS grid number, day of the year, and week of the year. Vessel in this dataset was set to A, and daylight set to night for all points. This dataset was used to make prediction plots for the best model. Plots with predicted S'_A and standard error of the predictions were created to demonstrate a practical application of these data in identifying changes in favorable pollock habitat during the winter fishery, a period when other data on pollock distribution and dynamics are scarce.

Investigations into ascertaining the stability of the commercial acoustic systems are on-going. Some work has been done using regression models over different time periods or areas where the fishing vessels have fished in tandem to determine if the systems have changed, but this work is not yet completed. Examination of the raw data did not reveal any noticeable sudden changes, but gradual changes over time would not be noticed. That all three vessels show similar trends provides some indication that if there were changes to the acoustic system they were not substantial. We assume, for the purposes of this study, that the acoustic systems were stable for the duration of the fishery and that any changes were not significant to our results.

Results/Discussion

Data description

The opportunistically collected acoustic data is a unique data set with many complicating features that hinder standard analysis and modeling techniques. One of the more difficult issues to overcome is that the data were not collected randomly, but were collected along the vessels' paths as the vessels conducted fishing operations. The data were therefore not independent, exhibiting significant spatial and temporal autocorrelation, and biased by fishing practices. The spatial and temporal sampling coverage over the study area did not fit a standard survey plan, and were concentrated in areas where and when fishers found harvestable densities of pollock, with some coverage in non-favorable areas as fishers transited among known fishing grounds (Fig. 1). Vessels A and B collected data throughout the study period, while Vessel C stopped collecting data on day 40 and only resumed for a very short period on day 74. Apparent temporal gaps in data collection were due to Vessels A and B delivering catch at the same time.

QQplots of the S'_A show that they are somewhat over-dispersed (Fig. 2), but due to the large number of samples in our dataset, statistical tests for heterogeneity among distributions such as the Kolmogorov-Smirnov test are, although highly significant ($p < 0.001$), in practice not meaningful. For the purposes of this study we assume the data are normally distributed, even though this is not strictly the case. The mean S'_A differ among vessels and between day and night. S'_A appears heteroscedastic for all vessels (Table 1; Fig. 3) with higher variance in the daytime consistent with aggregations forming dense, well-defined structures during the daytime and loose ill-defined structures at night.

Temporal and spatial explorations

The temporal analysis using ACF with data binned at one hour reveal significant periodicity in the distribution of S'_A at scales consistent with a diel cycle (Fig. 4) for all three vessels. The strength of the diel cycle in the data made it difficult to investigate other temporal trends at this scale. For data binned at 24-hours Vessels A and B show an apparent, although not significant at $\alpha=0.05$, gradual shift from positive to negative correlation for points at increasingly distant times (Fig. 4). Vessels B and C show a consistent pattern of auto-correlation for the first 40 days.

We calculated both daytime and nighttime cross-correlograms for each vessel for spatial bins of 0.5, 1 and 10 km. Only the 0.5 km cross-correlograms are presented since no additional insight was gained by increasing bin size. Differences in the spatial structure of pollock aggregations between day and night is evidenced by a lower correlation and steeper initial slope between 0.5 and 5 km in the daytime data , but a variable zero intercept ranging between 6 km and 15 km (Fig. 5). At night the slope of the cross-correlogram is a smooth curve between 3 and 13 km with an x-intercept between 9 and 11 km for all three vessels. When the day and night data are combined the x-intercept is at 10 km for all vessels. Nighttime data for all three vessels show significant periodicity with a wavelength of approximately 10 km to 20 km. Periodicity in the daytime data is not as distinct. Significant negative and then positive correlation between 150 km and 200 km observed for all vessels corresponds to the gap between southern and northern fishing zone. The irregularity of the daytime correlograms at scales less than 20 km may indicate spatial structure at a range of scales that overlap with inconsistencies in size and inter-aggregation distances, while the regular periodicity of the nighttime data suggest a consistent spatial structure.

Handling large unbalanced datasets

The CV_{RMSE} analysis revealed that inverse density biased sampling outperforms simple random sampling at sample sizes greater than 10,000 points (Fig.6). The analysis also suggests that an IDBS sample size of 20,000 points adequately represents the data with a CV_{RMSE} of 10%. A comparable random sample would require 35,000 data points. The IDBS resulted in datasets less influenced by dense data clustering on the fishing locations (Fig. 7) with a more even dispersion of sample points in time and space than random samples of the same size (Fig. 8).

Standardization and environmental covariates

The full model (Model 6) had the lowest AICc, including vessel and daylight as fixed effects, the IDBS strata as a random effect, and sea surface temperature and depth as non-linear processes with isotropic tensor splines. The model also included both an exponential spatial covariance structure and heteroscedastic variance between nighttime and daylight. In the initial set of candidate models there were large differences in AIC_c with a change of more than 2,000 between the best and the next best configuration (Table 2) indicating that all of the covariates included in the model are important factors influencing the distribution of observed S'_A.

Model 10 with a basis dimension of 10 for all smoothes had the highest log-likelihood in 69.5% of the cross-validation iterations (Table 3). The differences among the cross-validated models are marginal with changes in the prediction standard error ($\hat{\sigma}$) of less than 0.15 from the worst to best fitting configuration. QQplots of the standardized residuals from Model 10 show some over-dispersion (Fig. 10), as would be expected from modeling over-dispersed data and assuming a Gaussian distribution.

All model configurations show vessel as a significant effect with consistent differences among the vessels. Results from Model 10 suggest an S_A correction factor between Vessel A and B of -1.139 dB re 1 m^2nmi^{-2} and an S_A correction between Vessel A and C of -3.352 dB re 1 m^2nmi^{-2}. Although the models did find significant differences among the vessels the precision of these estimates are low with a 95% confidence range of between -0.525 and -1.752 dB re 1 m^2nmi^{-2} for Vessels A and B and -2.470 and -4.234 dB re 1 m^2nmi^{-2} for Vessels A and C (Table 5; Fig.11A). When translated into linear units (s'_A) the apparent pollock abundance observed by Vessel B was predicted to be 23% less than that observed by Vessel A with a 95% confidence range between 11% and 33% while Vessel C was predicted to have observed 54% less pollock abundance than Vessel A with a 95% confidence range between 43% and 62%.

The predicted S'_A at night was significantly higher than that observed in the daytime in all model configurations (Fig.11B). Model 10 estimates of the daylight effect suggests a 7.357 dB re 1 m^2nmi^{-2} increase in S'_A from daylight to night with a 95% confidence interval between 6.940 and 7.776 dB re 1 m^2nmi^{-2}. In linear terms the model predicts there to be 407% more fish available to acoustics at night than during the day with a 95% confidence range between 105% and 1150%. Model 10 also suggests that the variance of the S'_A between daylight and night is heteroscedastic with a more than halving of the standard deviation at night (variance correction factor $\delta_{night} = 0.52$; $CI_{95\%}$ between 0.53 and 0.50).

All models with SST included show a significant relationship between S'_A and SST (Fig.12A). There is an apparent threshold at 2°C, below which pollock density drops precipitously. At temperatures above 2°C up to the data maximum at 6.5°C, S'_A increases with temperature at a slower rate. An upper temperature threshold is not apparent from these data. Model 10 suggests some fluctuations in S'_A above 2°C with small peaks at 3.5°C and 5°C. The S'_A was also found to be related to bottom depth in all models that included the factor, with pollock densities highest between the 95 and 115 meter isobaths (Fig.12B). This peak in abundance near the 110m isobath coincides with observations on pollock distribution made during the summer acoustic and bottom trawl surveys (Acuna and Kotwicki, 2004.; Honkalheto et al, 2008).

Pollock abundance on the eastern Bering Sea shelf fishing grounds declined over the duration of the fishing season in all models, but with intermittent increases in abundance in models with higher flexibility (k > 6) in the week of the year smooth function. Less complex models, such as Model 6, fit with stiffer smoothes (k = 4) show a one-direction decline until week 9 (Fig.12C), while Model 10 found increases in abundance at weeks 6 and 8, but with an overall decline until week 9. After week 9 all models show an increase in abundance, but remaining well below the values observed in the start of the season. One issue that may not be fully addressed in the modeling and subsampling of these data is that at times the distribution of data points is sparse leading to the model potentially fitting to small spatial scale anomalies. The apparent temporal fluctuations in S'_A may in reality have been the vessels moving to areas of higher pollock density. Although Model 10 may have a better fit to the data, it may not have adequately represented actual temporal trend of pollock density fluctuations over the entire shelf. A less flexible model would smooth over the small-scale anomalies as observed in the smooth results from Model 6 (Fig. 12).

Daily projections of S'_A were predicted for the eastern Bering Sea shelf for 20 January to 1 April 2003 using both Model 10 (Fig. 13) and Model 6 (Fig. 14). These projections reveal a dynamic pollock habitat with S'_A fluctuating over the optimum depth range and time both with water temperatures and with the weekly decline in pollock abundance. The pattern of standard errors is as expected with low values over the shelf areas where and when the majority of data were collected and in the depth, temperature, and date ranges covered by the fishing vessels (Fig. 15). The model projects high standard error values for some regions under the ice in temperature ranges that were not sampled by the fishing fleet. As expected the standard error also increased when we attempt to project data outside of the date ranges sampled by the vessels.

Conclusion

The application of opportunistic acoustic data in traditional scientific studies is highly problematic because of the large volume of data collected, the lack of system and inter-vessel calibrations, and the absence of a predetermined sampling design. Although these data are not of the highest quality and must be used with caution and an understanding of their limitations, we have shown that they can be used quantitatively where absolute estimates of abundance are not necessary.

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Tables

Table 1: Mean ($\overline{S'_A}$) and standard deviation (σ) of the uncalibrated nautical area scattering strength (S'_A; dB re1m^{2}nm^{-2}) data for all data, for day and night, and for all vessels.

	$\overline{S'_A}$	Daytime $\overline{S'_A}$	Night $\overline{S'_A}$	Daytime σ of S'_A	Nighttime σ of S'_A
All	20.07	16.32	23.21	9.46	7.22
Vessel A	20.49	16.45	24.32	9.27	7.13
Vessel B	19.66	16.09	22.52	9.83	7.63
Vessel C	19.55	16.34	21.41	9.22	5.90
Number	290,033	132,136	157,897	132,136	157,897

Table 2: Configuration and results from initial model fitting, with small sample Akaike information criterion (AIC$_c$), and the change in AICc from the best fit model (Δ AIC$_c$). All models included vessel and daylight as fixed variables, Y or N indicates whether the covariate or variance/correlation structures were implemented. The best fit model is shaded.

M #	SST	Covariates Depth	Week	VAR	COR	AIC$_c$	Δ AIC$_c$
1	N	N	N	N	N	128810.9	5345.1
2	Y	N	N	N	N	128655.0	5189.2
3	Y	Y	N	N	N	128638.9	5173.1
4	Y	Y	Y	N	N	128597.4	5131.6
5	Y	Y	Y	Y	N	125548.8	2083.0
6	Y	Y	Y	Y	Y	123465.8	0.0

Table 3: Results from final cross-validation model selection. With the basis dimension (k) for the smooth term for a particular variable, effective degrees of freedom associated with the smooth (edf), the mean residual sum of squares (Mean RSS), the estimated standard error of the prediction ($\hat{\sigma}$), and the proportion of times the model configuration resulted in the highest prediction log-likelihood (Chosen Prop.).

M	SST		Depth		Week		Mean		Chosen
#	k	edf	k	Edf	k	edf	RSS	$\hat{\sigma}$	Prop.
7	3	1.983	3	1.937	3	1.965	64892.35	8.07	0.001
6	4	2.936	4	2.805	4	2.884	64358.91	8.02	0.001
8	6	4.340	6	3.985	6	4.762	63896.92	7.99	0.000
9	8	6.243	8	6.216	8	6.201	63266.70	7.95	0.066
10	10	7.599	10	6.916	10	7.972	62672.48	7.92	0.695
11	11	8.162	10	6.912	10	7.993	62679.78	7.92	0.237

Table 4: Model 10 results

Parametric coefficients:

	Estimate	Std. error	t-value	Pr(>|t|)
(Intercept)	15.756	0.265	59.517	< 0.0001
VESSELB	-1.139	0.312	-3.646	0.000267
VESSELC	-3.352	0.449	-7.461	8.98E-14
DAYLIGHT -- NIGHT	7.358	0.213	34.536	< 0.0001

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(SST)	8.264	8.764	17.329	< 0.0001
s(DEPTH_2)	6.069	6.569	8.494	6.53E-10
s(WEEK)	8.259	8.759	15.931	< 0.0001

R-sq.(adj)=	0.736
Scale est.=	87.401
n =	19,513

Table 5: Approximate 95% confidence intervals for Model 10 coefficients and variance parameters.

Covariate	Lower	Estimate	Upper
Fixed Effects:			
Intercept	15.237	15.756	16.275
Vessel B	-1.752	-1.139	-0.525
Vessel C	-4.234	-3.352	-2.470
Daylight - Night	6.940	7.358	7.776
Range of Smooth term effects:			
SST	-5.902	-1.115	3.672
Depth	-1.763	0.243	2.249
Week	3.127	7.048	10.969
Random Effects:			
IDBS Grid sd(intercept)	4.213	4.439	4.677
Variance Function:			
Daylight - Night	0.505	0.515	0.526
Within Group Standard Error	9.202	9.349	9.498

Figures

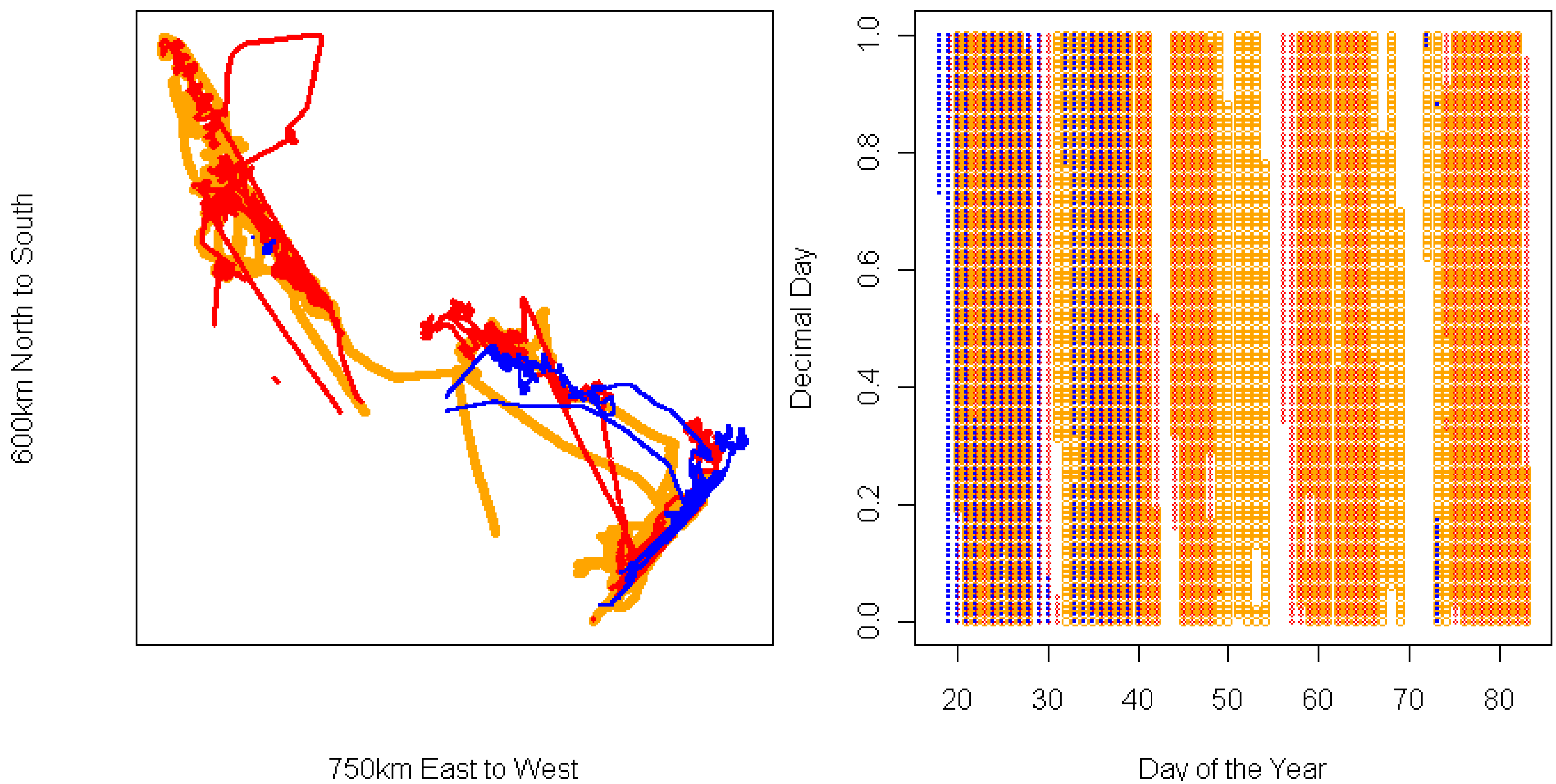

Figure 1: Relative spatial and temporal distribution of opportunistically collected acoustic data from three vessels (A – orange, B – red, and C – blue) participating in the eastern Bering Sea pollock fishery 18 January 2003 to 26 March 2003.

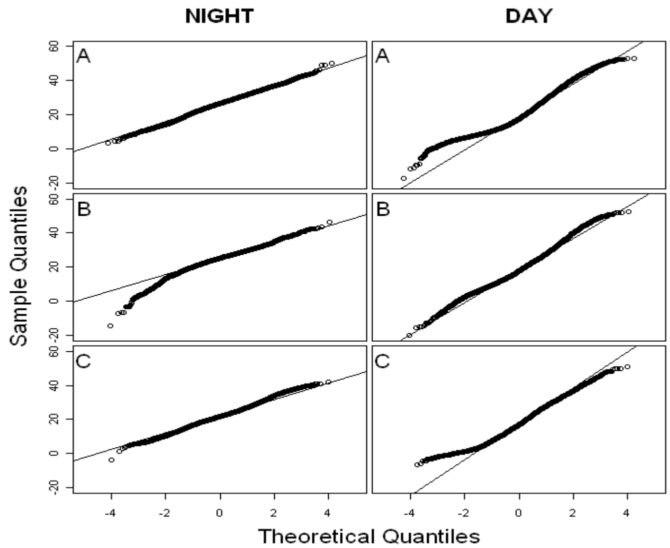

Figure 2: QQplots of the nautical area scattering strength (S_A ; dB re $1(m^2nm^{-1})$) from all three vessels by day and night.

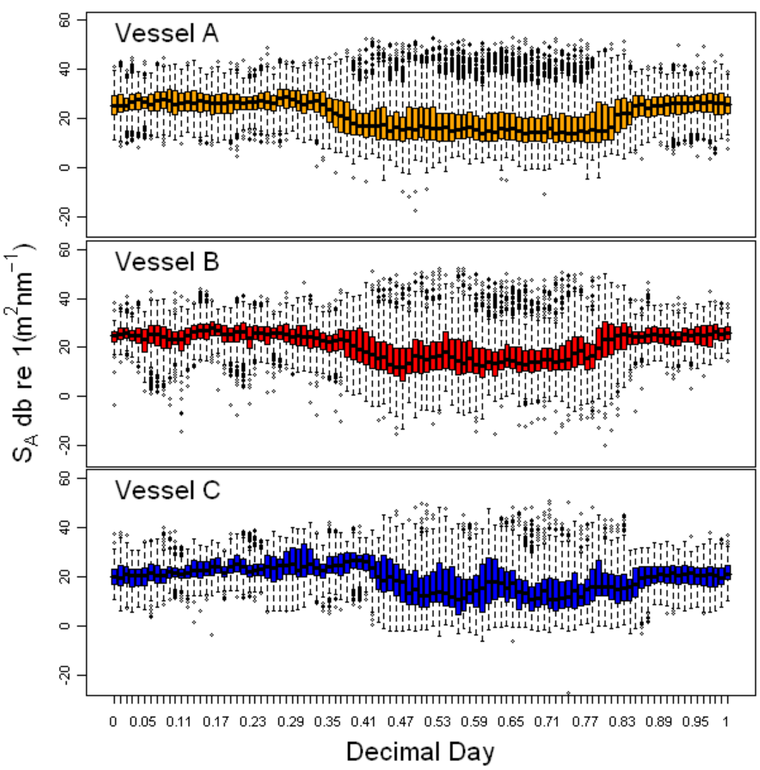

Figure 3: Boxplot of nautical area scattering strength (S_A ; dB re $1(m^2nm^{-1})$) over decimal time of day.

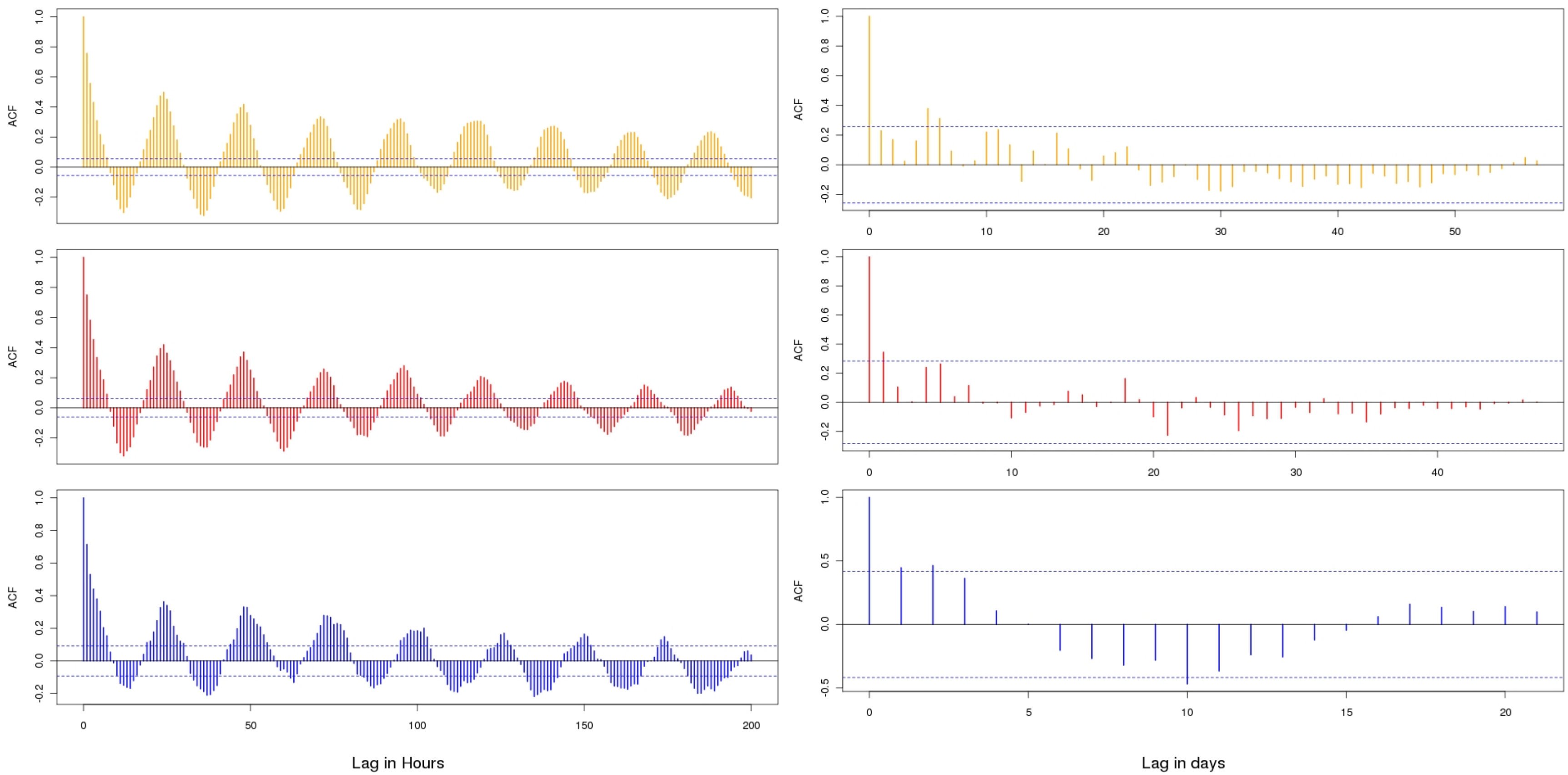

Figure 4: Autocorrelation function plots of nautical area scattering strength (S_A ; dB re $1(m^2nm^{-1})$) for data binned by 1-hour (left) and 24-hour(right) for the three vessels (A , top; B, middle; C, bottom), dashed lines are the significance levels (α =0.05). Note the x-axis in the final 24-hour binned graph differs for the others.

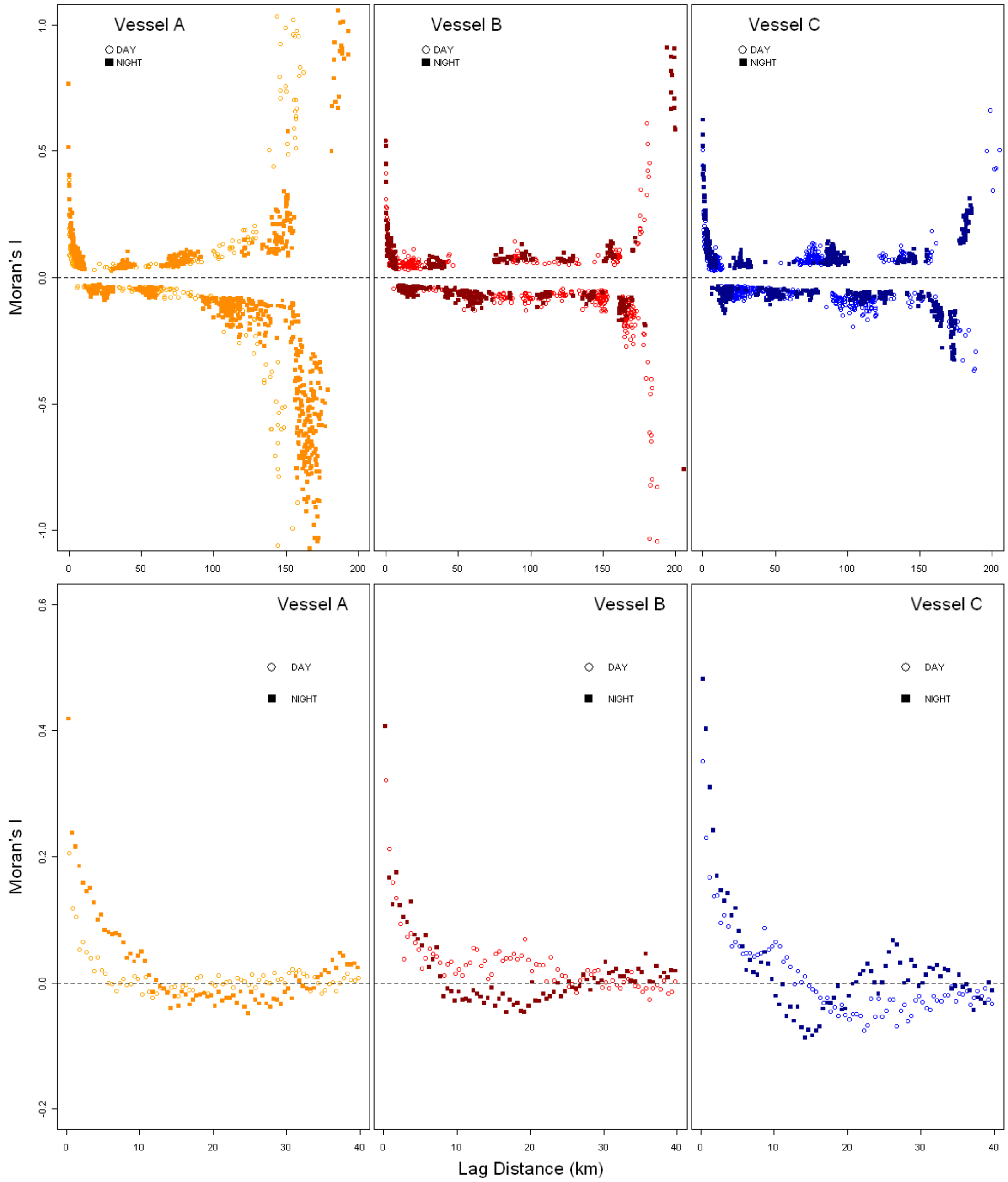

Figure 5: Moran's I cross-correlograms of the nautical area scattering strength (S_A ; dB re $1(m^2nm^{-1})$) at 1km resolution for day and night and all three vessels for lags out to 200km (top) non-significant correlations are not included in the figure and 40 km (bottom).

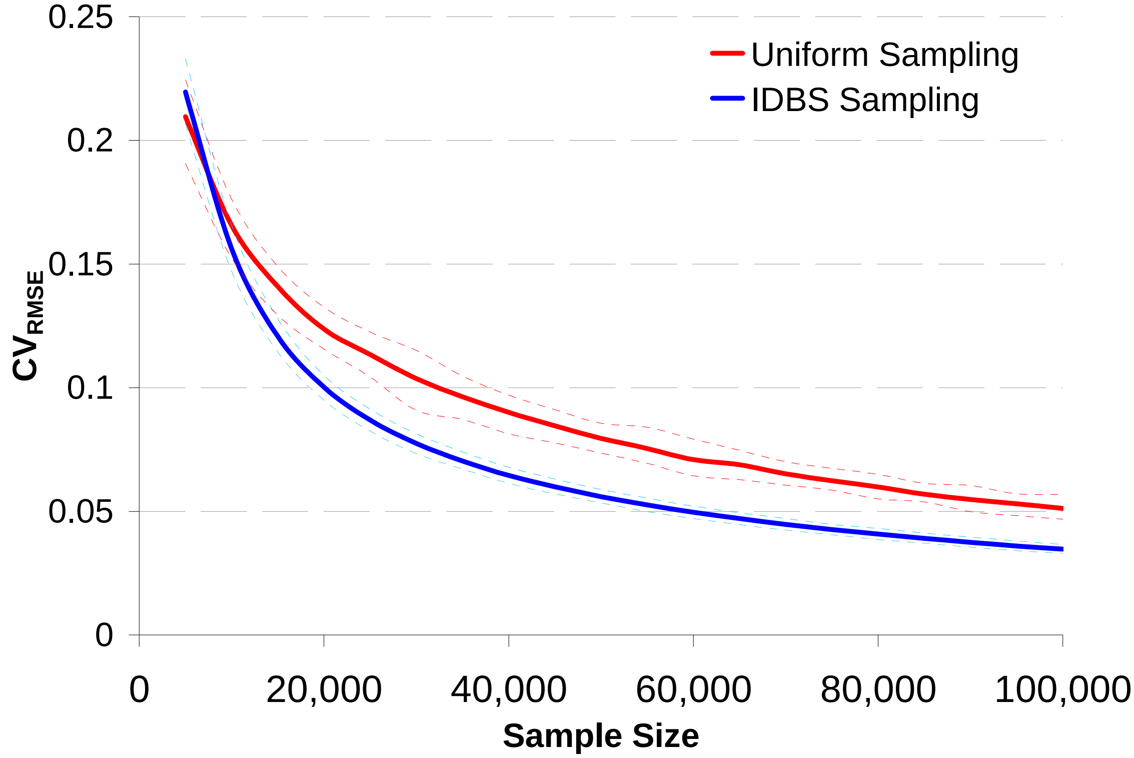

Figure 6: CV_{RMSE} analysis of the inverse density bias sample size showing both IDBS and uniform samples over sample sizes of 5,000 to 100,000. The dashed lines are bootstrapped 95% confidence bounds for the estimates over 1,000 iterations.

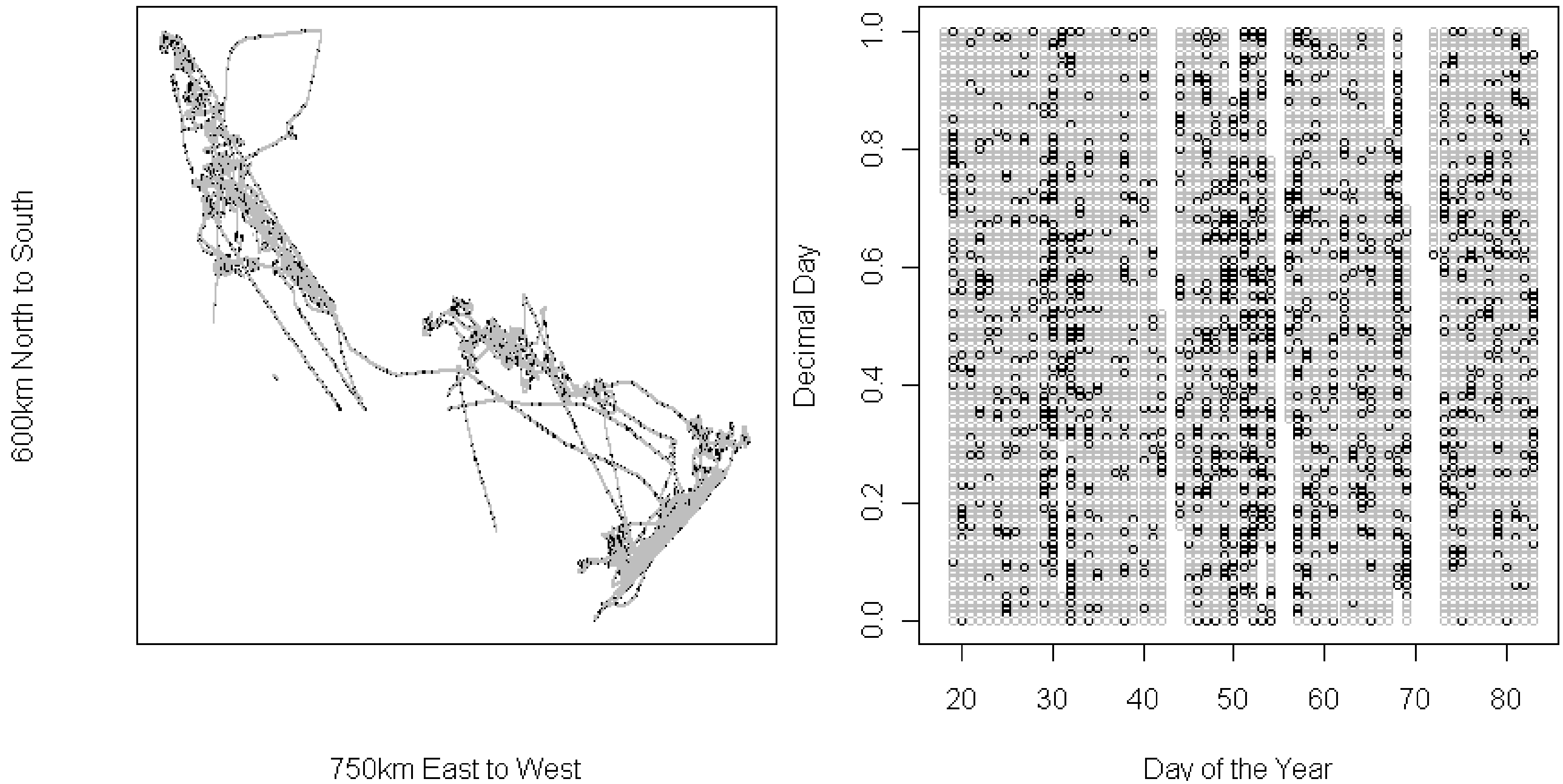

Figure 7: Spatial (left) and temporal (right) Coverage of IDBS with sample size of 20,000, underlying black points are all of the data overlying grey points are from the IDBS sample, a visible black point means it was not in the subsample.

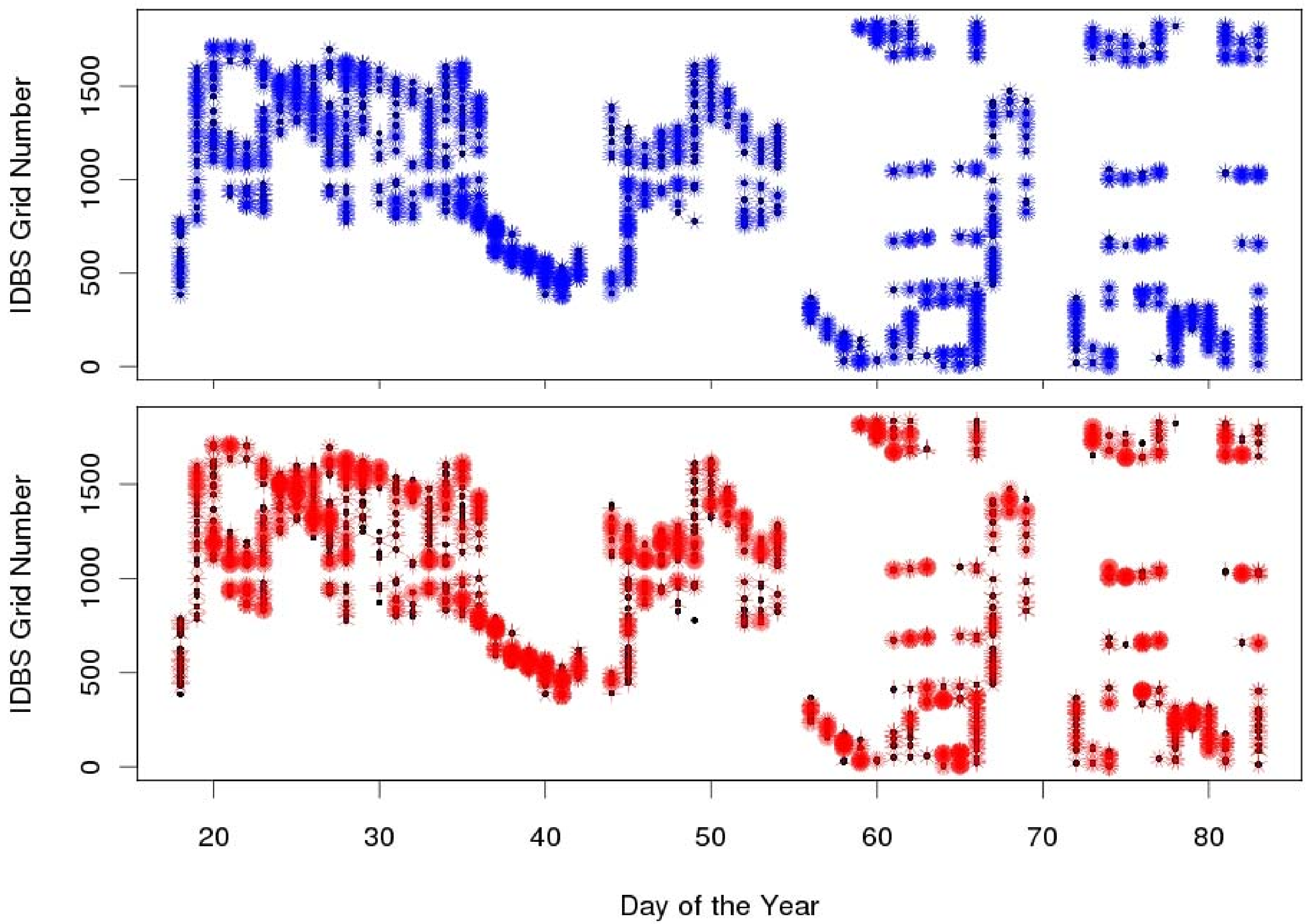

Figure 8: Plot of inverse density biased (top) and uniform random (bottom) subsamples of 20,000 points by day of the year and IDBS grid number, number of leaves on the diagram indicate the number of samples for the grid cell.

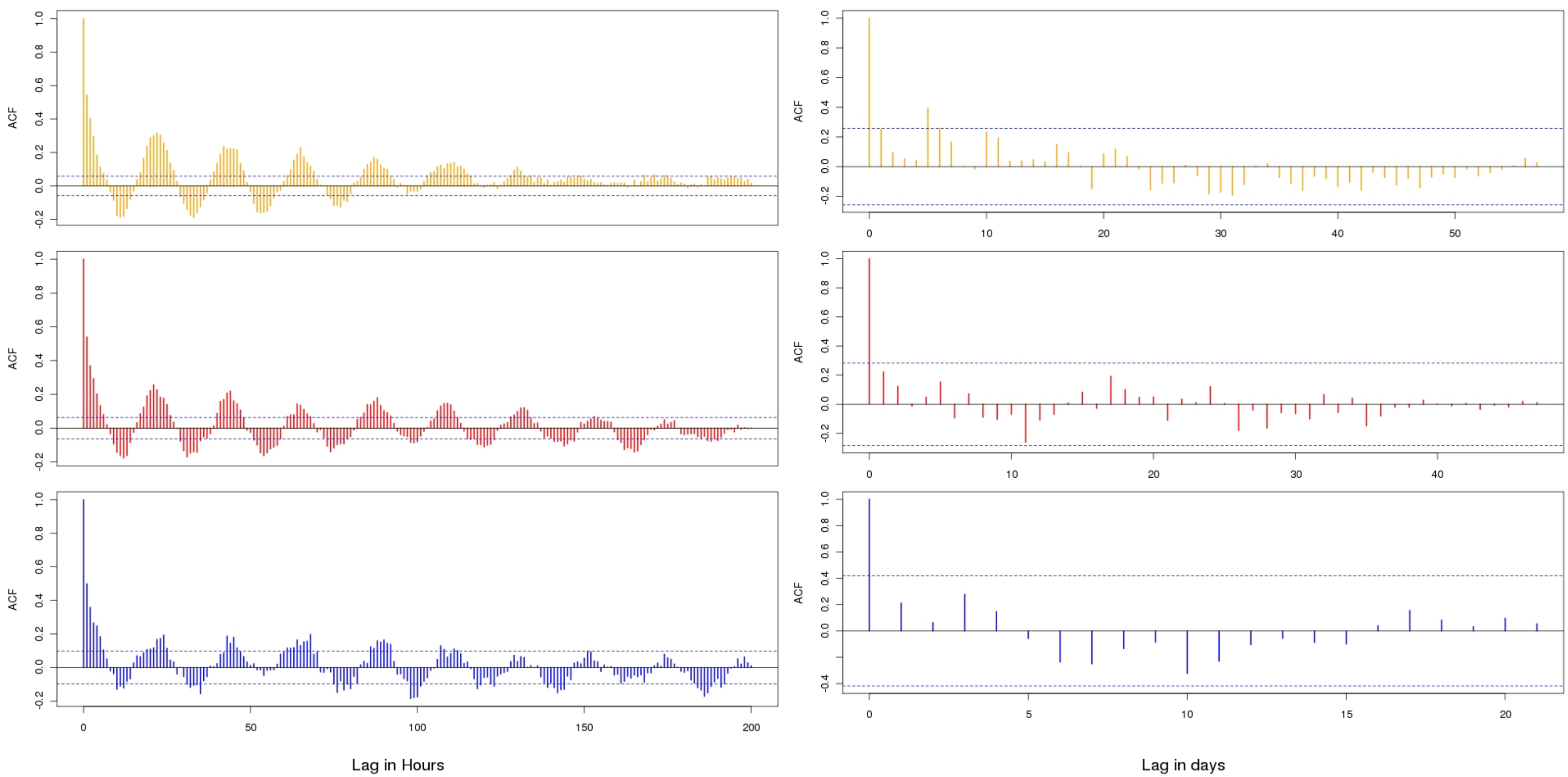

Figure 9: Auto-correlation function (ACF) plots of nautical area scattering strength (S_A ; dB re $1(m^2 nm^{-1})$) in hour (left) and day (right) lags from all three vessels for the inverse density biased sampling, dashed lines are the significance levels (α =0.05).

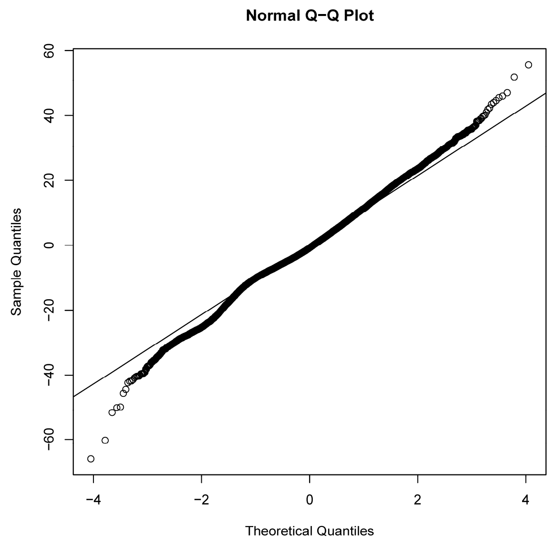

Figure 10: QQplot of Pearson's residuals for Model 10

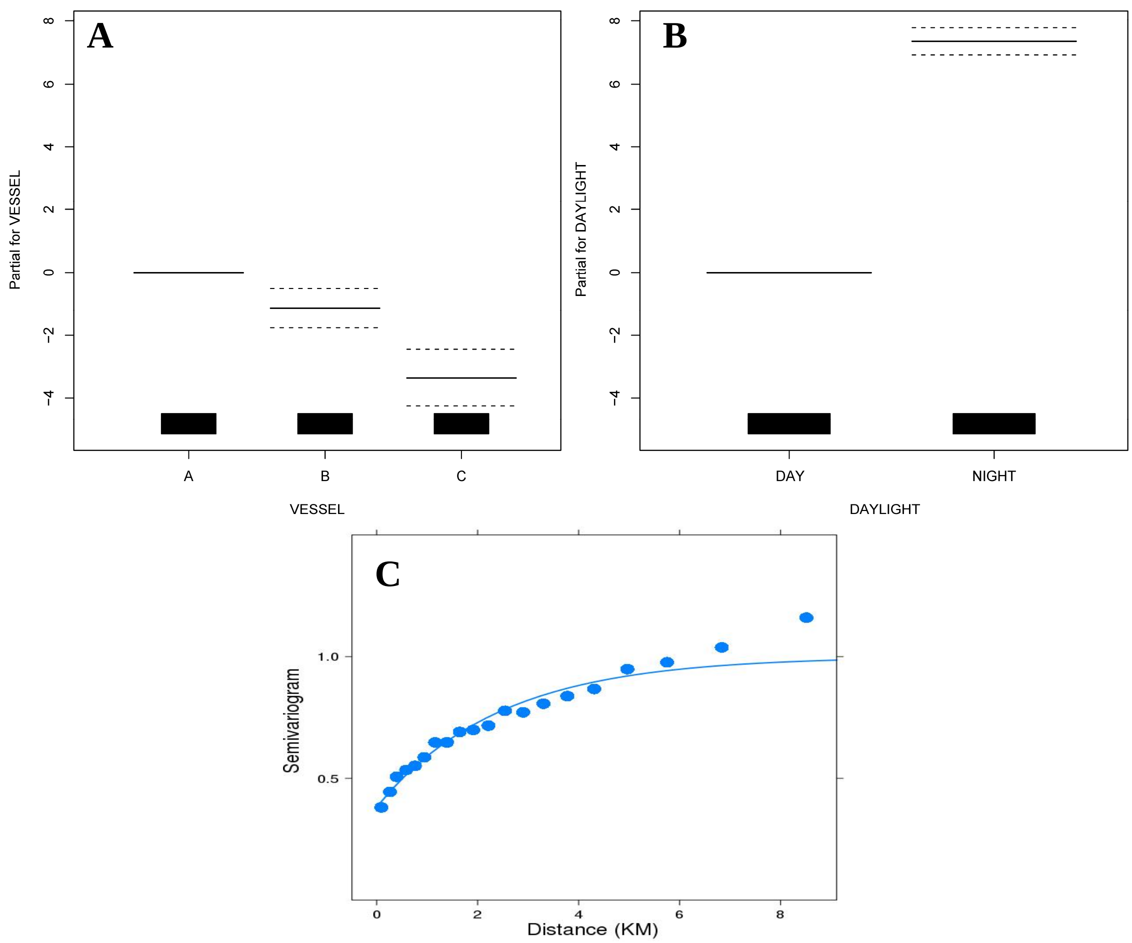

Figure 11: Model 10 results: A) difference in S'_A measurements among vessels, B) difference in uncalibrated nautical scattering strength S'_A from night to day, and C) fit of the data to an exponential variogram with a range of 2.42 and nugget of 0.382.

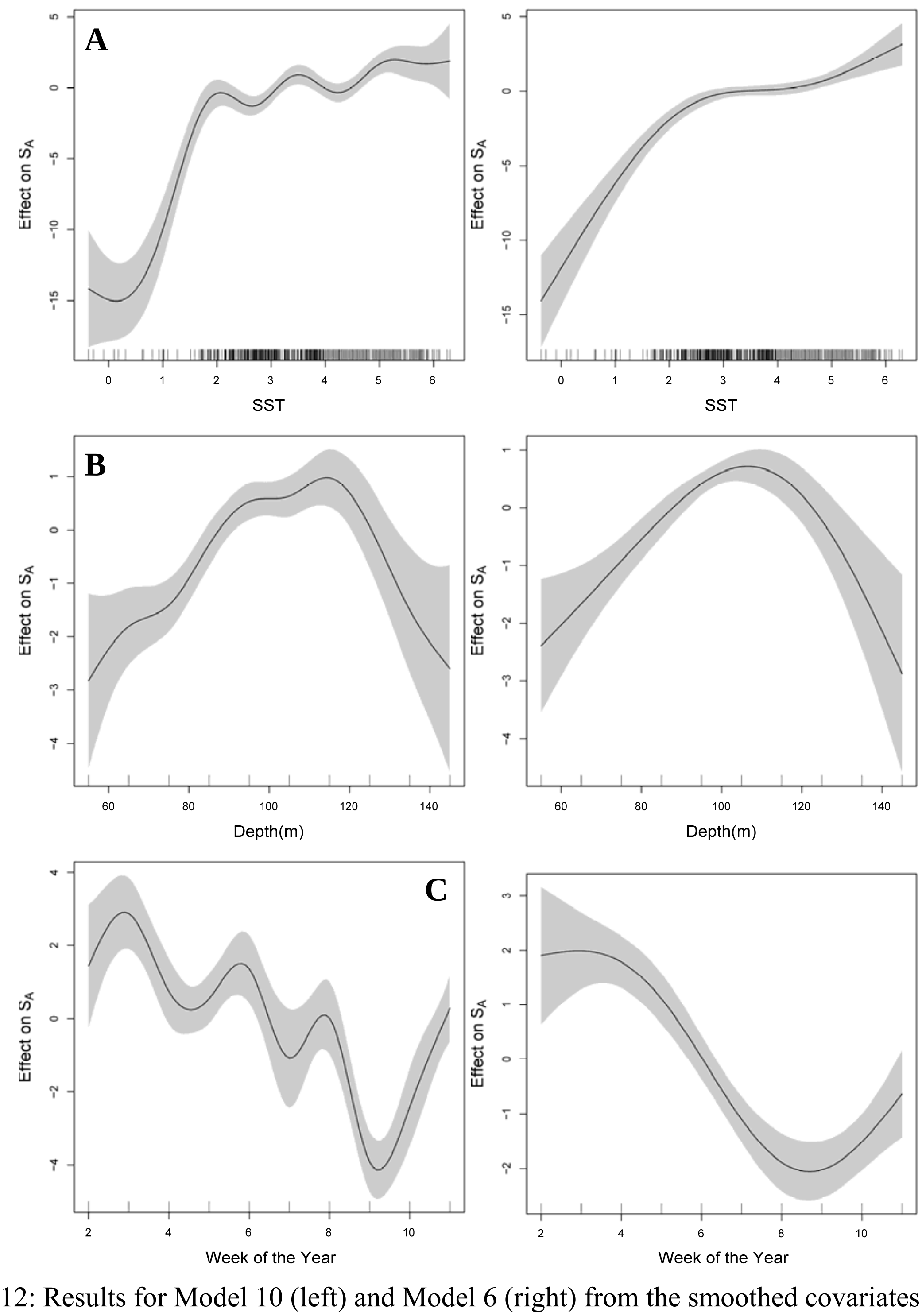

Figure 12: Results for Model 10 (left) and Model 6 (right) from the smoothed covariates showing: A) sea surface temperature, B) bottom depth (m), and C) week of the year effects on uncalibrated nautical scattering strength S'_A. Note the y-axis scales differ.

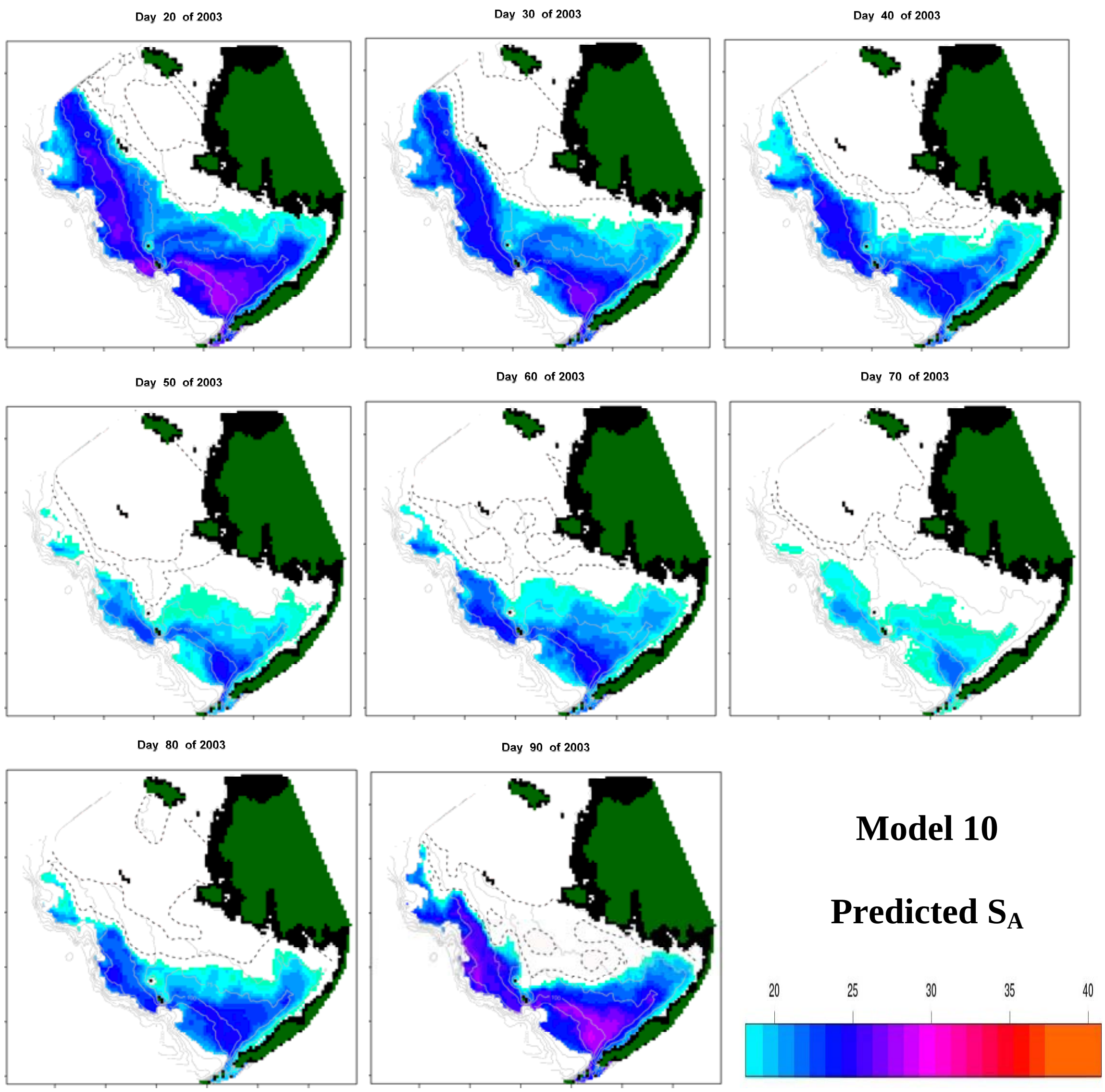

Figure 13: Projection of uncalibrated nautical area scattering strength (S'_A; dB re 1 m^{2}nmi^{-2}) from Model 10 for 10-day increments starting 20 January 2003 through 1 April 2003. S'_A is scaled from red highest to light blue lowest, dotted lines are the 1° and 0° C SST isotherms, solid grey lines are bathymetric contours, blacked out areas are depths less than 25 m.

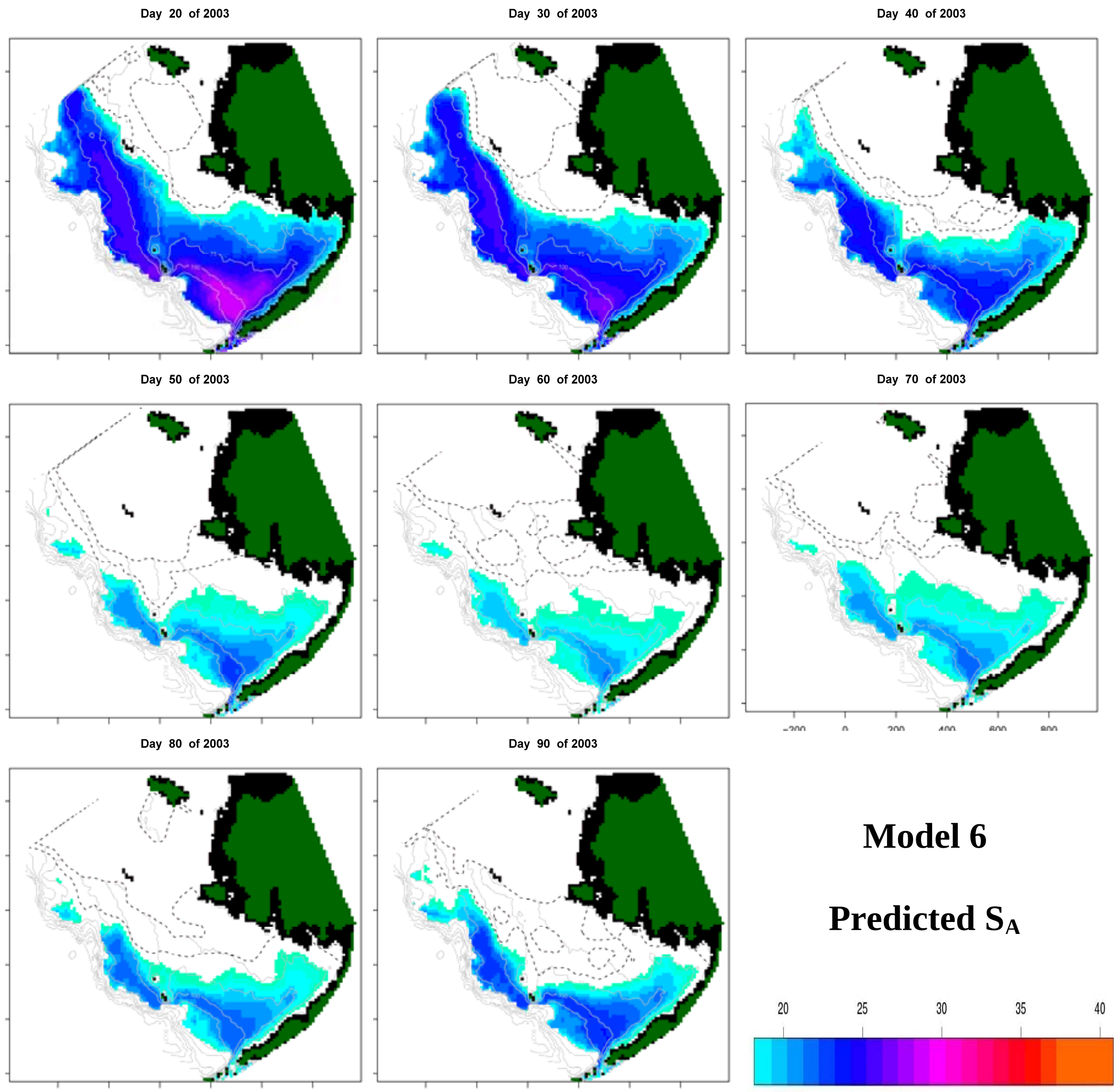

Figure 14: Projection of uncalibrated nautical area scattering strength (S'_A; dB re 1 m^{2}nmi^{-2}) from Model 6 for 10-day increments starting 20 January 2003 through 1 April 2003. S'_A is scaled from red highest to light blue lowest, dotted lines are the 1° and 0° C SST isotherms, solid grey lines are bathymetric contours, blacked out areas are depths less than 25 m.

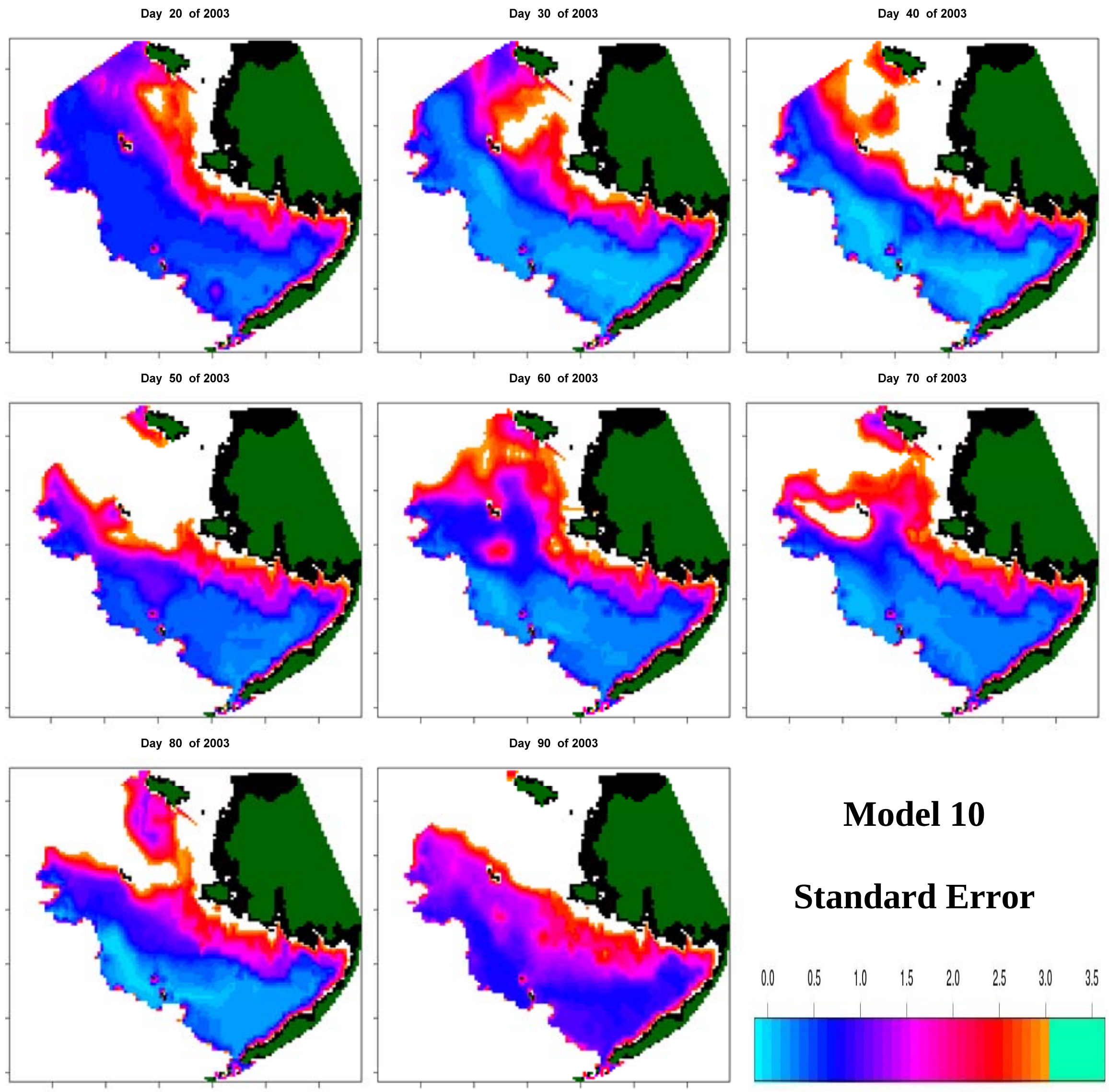

Figure 15: Projection of the prediction standard error from Model 10 for 10-day increments starting 20 January 2003 through 1 April 2003. Blacked out areas are depths less than 25 m.